FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A particle is projected at an angle θ = 30º with the horizontal, with a velocity of 10 m/s then
   (A) After 2 s the velocity of particle makes an angle of 60º with initial velocity vector
   (B) At 1 s the velocity of particle makes an angle of 60º with initial velocity vector
   (C) The magnitude of velocity of particle after 1 s is 10 m/s
   (D) The magnitude of velocity of particle after 1 s is 5 m/s

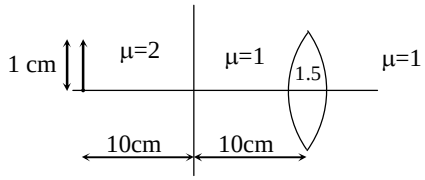
2. An electron in H-atom jumps from second excited state to first excited state and then from first excited to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases be *a, b* and *c* respectively. Then
   (A) $c = \frac{1}{a}$ (B) $a = \frac{9}{4}$
   (C) $b = \frac{5}{27}$ (D) $c = \frac{5}{27}$

3. If $\vec{A} \times \vec{B} = \vec{C} + \vec{D}$, then select the correct alternative(s).
   (A) $\vec{B}$ is perpendicular to $\vec{C} + \vec{D}$
   (B) $\vec{A}$ is perpendicular to $\vec{C}$
   (C) Component of $\vec{C}$ along $\vec{A}$ = component of $\vec{D}$ along $\vec{A}$
   (D) Component of $\vec{C}$ along $\vec{A}$ = – component of $\vec{D}$ along $\vec{A}$

4. A vibrating string produces 2 beats per seconds when sounded with a tuning fork of frequency 256 Hz. increasing the tension in the string produces 3 beats per second. The initial frequency of the string may have been
   (A) 253 Hz (B) 254 Hz
   (C) 258 Hz (D) 259 Hz

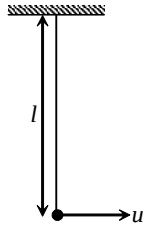
Space for Rough work

5. An object of length 1m is placed on a principle axis of biconvex lens of radius 5 cm. Distance between the lens and object is 20 cm. Space between the lens and object is filled with medium of two different refractive index 2 and 1 as shown in the figure. Refractive index is 1 on the left of the object and on the right side of the lens. Boundary of both medium is mid-way between the object and lens as shown in figure.



(A) The image will be formed at distance of 7.5 cm from the optical centre.
(B) The image will be formed at distance of 10 cm from the optical centre.
(C) The size of the image is 0.5 cm.
(D) The size of the image is 0.4 cm.

6. A particle of mass m is attached to one end of the light inextensible string and other end of the string is fixed in vertical plane as shown. Now particle is given horizontal velocity $u = \sqrt{\frac{5}{2}gl}$, then which of the following statement/s is/are correct.



(A) The string makes angle with the downward vertical is 60^0 when acceleration of the particle is horizontal.
(B) The tension in string is 2 mg when acceleration of the particle is horizontal.
(C) The speed of the particle is $\sqrt{\frac{3}{2}gl}$ when acceleration of the particle is horizontal
(D) The particle will not complete the circle.

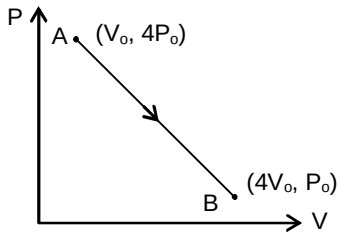
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(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

Consider a monoatomic ideal gas undergoing a process AB as shown in figure.



7. Find the maximum temperature of the gas if temperature at A is T_o.

(A) $\frac{21}{16}T_o$ (B) $\frac{23}{16}T_o$

(C) $\frac{25}{16}T_o$ (D) $\frac{27}{16}T_o$

8. Volume at which the process becomes exothermic from endothermic is

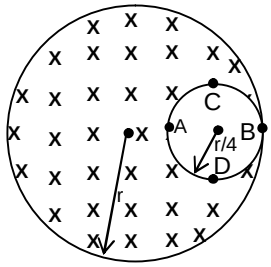
(A) $\frac{19}{8}V_o$ (B) $\frac{21}{8}V_o$

(C) $\frac{23}{8}V_o$ (D) $\frac{25}{8}V_o$

Space for Rough work

Paragraph for Question Nos. 9 and 10

Consider a spherical charge distribution of volume charge density ρ c/m^3



9. Potential difference between A & B

(A) $\frac{2\rho}{3\varepsilon_0} r^2$ (B) 0

(C) $\frac{\rho}{4\varepsilon_0} r^2$ (D) $\frac{\rho r^2}{8\varepsilon_0}$

10. Potential difference between C & D

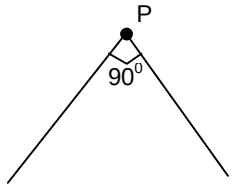
(A) 0 (B) $\frac{2\rho_r}{3\varepsilon_0}$

(C) $\frac{\rho_r}{4\varepsilon_0}$ (D) none of the above

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

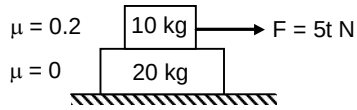
11. A system of two identical rods (L shaped) of mass m, length L. System rest on a peg p as shown in the figure. Period of oscillations $T = 2\pi\sqrt{\frac{n\sqrt{2}\,L}{3g}}$ then n = ?



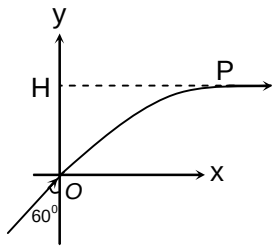
Space for Rough work

12. A monkey of mass m is sitting on a platform of mass M. Monkey can jump with a velocity of 5 m/s making an angle 37^0 with the horizontal with respect to platform. To jump the monkey 1 meter with respect to the ground. Find out the value of m/M.

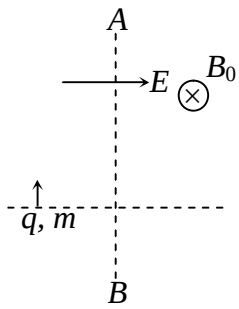
13. What is the speed of 10 kg block at the end of 3 second.



14. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \le y \le 1$. A ray of light is incident at origin at an angle 60^0 with *y* –axis as shown in the figure. At point P ray becomes parallel to x-axis. Find the value of 'H'.



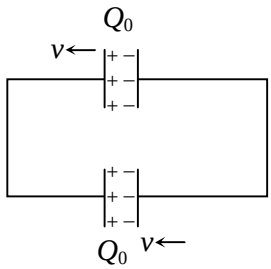
15. A uniform electric field *E* is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line *AB* as shown. A charge particle having charge *q* and mass *m* is projected vertically upward and crosses the line *AB* after time t_0. The speed of projection if particle moves after t_0 with constant velocity. (given $qE = mg$) is nt_o. Find the value of 'n'. (g = 9.8 m/s^2)



16. The refractive index for water w.r.t. air for a sound wave if the velocities of sound in water and air are 1440 ms^{-1} and 340ms^{-1}, is approximately,

17. When entire Young's double slit apparatus is immersed in a liquid the fringe width decreases by 20%. The refractive index of the liquid is K×0.25. Then find the value of K.

18. Two identical capacitor connected as shown and having initial charge Q_0. Separation between plates of capacitor is d_0. Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed *v* towards left while other plate of capacitor remains fixed. (given $\frac{Q_0 v}{2d} = 1$ amp). Find the value of current (in amp) in the circuit.



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. 30 mL of CH_3OH(d = 0.8 g/cc) is mixed with 60 mL of C_2H_5OH (d = 0.92 g/cc) at 25°C to form a solution of density 0.88 g/cc.
Choose the correct statement(s) from the following.
(A) Molarity of the solution and percentage change in volume are 13.59 and zero respectively.
(B) The mole fraction of solute and molality are 0.385 and 13.59 respectively.
(C) The mole fraction of solvent and molality are 0.615 and 13.59 respectively.
(D) Molarity and molality of resulting solution are 6.33 and 13.59 respectively.

20. Which of the following statement(s) is/are correct for $Na_3[Co(ONO)_6]$?
(A) It's aqueous solution is slightly basic in nature.
(B) It's IUPAC name is sodium hexanitrito cobaltate(III)
(C) It is called sodium cobaltinitrite
(D) One molecule of it forms three ions in water.

21. The total vapour pressure of a binary solution is given by
$P = (80\ X_A + 240\ X_B)$ mm Hg
Where X_A and X_B are the mole fractions of components A and B. Choose the correct statement(s) from the following:
(A) vapour pressure of solution is less than that of the pure B component.
(B) vapour pressure of solution is more than that of pure A component.
(C) vapour pressure of pure A is 80 mm of Hg and that of pure B is 240 mm Hg.
(D) the vapour pressure of pure A and B are 240 and 80 mm Hg respectively.

22. Which of the following cations will turn a borax bead blue in an oxidizing flame?
(A) Fe^{3+} (B) Fe^{2+}
(C) Co^{2+} (D) Cu^{2+}

Space for Rough work

23. Consider the following electrodes with their E^o values:

$E^o_{Cu^{2+}/Cu} = +0.34\,V$, $E^o_{Ni^{2+}/Ni} = -0.25\,V$

$E^o_{Ag^{+}/Ag} = +0.8\,V$, $E_{Zn^{2+}/Zn} = -0.76\,V$

Which of the following reaction(s) is/are spontaneous?

(A) $Cu(s) + Ni^{2+}(aq) \longrightarrow Ni(s) + Cu^{2+}(aq)$

(B) $Cu(s) + 2Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

(C) $Cu(s) + 2H^{+}(aq) \longrightarrow Cu^{2+}(aq) + H_2(g)$

(D) $Zn(s) + 2H^{+}(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$

24. Which of the following are intensive properties?

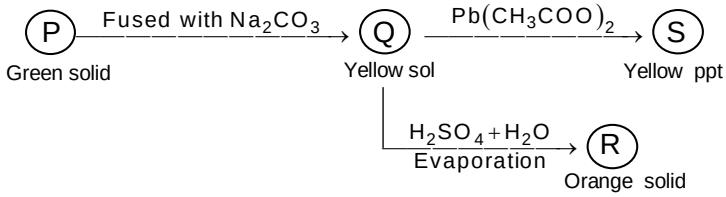
(A) pH (B) Molarity

(C) Refractive index (D) Entropy

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 25 and 26



Answer the following questions on the basis of above reaction series.

25. Which of the following is (P) ?

(A) $FeSO_4.7H_2O$ (B) $Cr_2(SO_4)_3$

(C) Anhy.$FeSO_4$ (D) CrO_5

Space for Rough work

26. The yellow solution (Q) is

(A) $FeCO_3$ (B) $Cr_2(CO_3)_3$
(C) Na_2CrO_4 (D) Na_4FeO_3

Paragraph for Question Nos. 27 and 28

Activity series
K
Na
Ca
Mg
Al
Zn
Fe
Pb
H
Cu
Ag
Au

The electrolysis of aqueous solution of an electrolyte is accompanied with deposition of metal or liberation of H_2 gas at cathode. A gas is also liberated at the anode. H_2 gas is liberated if the metal forming the cation of the electrolyte is present above hydrogen in the activity series. The metal will be deposited if it is present below hydrogen in the activity series. The mass of metal deposited at the cathode is directly proportional to the quantity of electricity passed through the electrolyte.

Answer the following questions on the basis of above write up.

27. The electrolysis of which of the following solution liberates H_2 gas at cathode and Cl_2 gas at anode?

(A) $CuCl_2 + H_2O$ (B) $NaCl + H_2O$
(C) $HgCl_2 + H_2O$ (D) $AuCl_3 + H_2O$

28. Which of the following is deposited or liberated at cathode if aqueous solution of potash alum ($K_2SO_4.Al_2(SO_4)_3.24H_2O$) undergoes electrolysis?

(A) K (B) Al
(C) H_2 (D) O_2

Space for Rough work

SECTION – D
(Numerical Answer Type)

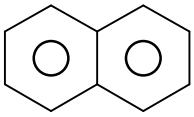
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. An octahedral complex of chromium contains NH_3 and Cl ligands. The oxidation number of chromium in the complex is +3. The complex shows one cis and one trans isomer(no fac or mer isomers). Reaction of the complex with $AgNO_3$ solution forms white precipitate. How much precipitate in gram will be formed if one mole of the complex reacts with excess of $AgNO_3$ solution?

30. HA is a weak acid and NaA is it's salt. An aqueous solution contains HA and NaA in the molar ratio of 10 :1 respectively. The pH of this aqueous solution is 5.38. What is the pK_a of the acid HA?

31. How much heat in Joule unit can be absorbed by heating 4 g of helium gas from 40^oC to 140^oC at constant pressure? $\left[C_P = \frac{5R}{2}\right]$

32. A polychloride(P) reacts with aqueous NaOH to form a monocarboxylic acid(Q). (Q) reduces Tollen's reagents and does not respond to H.V.Z reaction. What is the difference in molar mass between (P) and (Q) in gram unit?

33. 
$\xrightarrow[\text{Anhy. } AlCl_3]{CH_3Cl(1\,eq)} (X) \xrightarrow{ClF(1\,eq)} (Y)$
What is the molar mass of (Y) in g unit?

34. At a certain temperature the density of a metal is 1.5 g/cc. due to defect in the solid, the relative number of atoms per unit cell is found to be 0.03. Find the molar mass of the metal in gram unit if the edge length of its unit cell is $1\,\text{Å}$.[N = 6.022×10^{23}]

35. One mole of glycol was dissolved in 2 kg water and the resulting solution was cooled to -6^oC. How many gram of ice is separated in the process?
[K_f of H_2O = 1.86 K Kg mol^{-1}]

36. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
If above reaction takes place in a 100 L container at 1000 K, 100 moles of CO_2 is produced at equilibrium. What is the equilibrium constant K_P of the reaction in atm unit?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let z_1, z_2, z_3 be the coordinate of vertices *A*, *B*, *C* of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(A) $\text{Re}\left(\frac{W_2}{W_1}\right) = 1$ (B) $\text{Re}(W_1\bar{W}_2) = |W_1|^2$

(C) $\text{Re}(W_1\bar{W}_2) = |W_2|^2$ (D) $\text{Re}\left(\frac{W_1}{W_2}\right) = 1$

38. If $A(\alpha,\beta) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{\beta} \end{bmatrix}$, then

(A) $A(\alpha,\beta)' = A(-\alpha, \beta)$ (B) $A(\alpha, \beta)^{-1} = A(-\alpha,-\beta)$

(C) $\text{Adj}(A(\alpha,\beta)) = e^{\beta}A(-\alpha, -\beta)$ (D) $A(\alpha,\beta)' = A(\alpha,-\beta)$

39. If $\int_0^{\alpha} \frac{dx}{1-\cos\alpha\cos x} = \frac{A}{\sin\alpha} + B(a \neq 0)$. Then possible values of A and B are

(A) $A = \frac{\pi}{2}, B = 0$ (B) $A = \frac{\pi}{4}, B = \frac{\pi}{4\sin\alpha}$

(C) $A = \frac{\pi}{6}, B = \frac{\pi}{\sin\alpha}$ (D) $A = \pi, B = \frac{\pi}{\sin\alpha}$

Space for Rough work

40. The solution of the differential equation $\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y_1(0) = 1$ and $y(0) = 0$ is

(A) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$ (B) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x$

(C) $\frac{-\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{1}{3}x + 1$ (D) none of these

41. If θ is the angle subtended at $P(x_1, y_1)$ by the circle $S = x^2 + y^2 + 2gx + 2fy + c = 0$, then

(A) $\cot\theta = \frac{\sqrt{S_1}}{\sqrt{g^2 + f^2 - c}}$ (B) $\cot\frac{\theta}{2} = \frac{\sqrt{S_1}}{\sqrt{g^2 + f^2 - c}}$

(C) $\tan\theta = \frac{2\sqrt{g^2 + f^2 - c}}{\sqrt{S_1}}$ (D) $\theta = 2\tan^{-1}\left(\frac{\sqrt{g^2 + f^2 - c}}{\sqrt{S_1}}\right)$

where $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

42. If sines of the angle A and B of a triangle ABC satisfy the equation $c^2x^2 - c(a+b)x + ab = 0$, then the triangle

(A) is acute angled (B) is right angled

(C) is obtuse angled (D) satisfies $\sin A + \cos A = \frac{(a+b)}{c}$

Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Let $f(x)=1+\int_0^1\left(xe^y+ye^x\right)f(y)dy$ when x and y are independent variables.

43. If complete solution set of 'x' for which function $h(x)=f(x)+3x$ is strictly increasing $(-\infty, k)$ then $\left[\frac{4}{3}e^k\right]$ equals to: (where [.] denotes greatest integer function):

(A) 1 (B) 2
(C) 3 (D) 4

44. If acute angle of intersection of the curve $\frac{x}{2}+\frac{y}{3}+\frac{1}{3}=0$ and $y=f(x)$ be θ then $\tan\theta$ equals to:

(A) $\frac{8}{25}$ (B) $\frac{16}{25}$
(C) $\frac{14}{25}$ (D) $\frac{4}{5}$

Paragraph for Question Nos. 45 and 46

Consider the function $f(x)$, a fourth degree polynomial such that

$\lim_{x\to1}\frac{f(x)}{(x-1)^2}=1,\ f'(0)=-6, f'(2)=6$.

45. The minimum value of $f(x)$ is

(A) 0 (B) 1
(C) –1 (D) –2

Space for Rough work

46. The length of subtangent of the curve $y = f(x)$ where it cuts the y – axis is

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{1}{4}$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Four students appeared in three examinations. The number of possible ways in which we can have result, such that neither any students fails in all the examinations nor there is an examination which is not cleared by any student is k, then find $\frac{k}{100}$

48. If α is a complex seventh root of unity, then the equation whose roots are $\alpha + \alpha^2 + \alpha^4$ and $\alpha^3 + \alpha^5 + \alpha^6$ is $x^2 + x + c = 0$, then $\frac{22c}{7}$ is

49. The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points *A* and *B*, there is a point *P* on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2} - 1)$, the number of such points (*P*) is

50. If the line $3x + 4y = 12$ intersects the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$ at *P* & P' and its asymptotes at *Q* and *Q'* then value of $\frac{\frac{PQ}{P'Q'} + \frac{PQ'}{P'Q}}{100}$ is

Space for Rough work

51. Let $\vec{b} = 4i + 3j$ and $\vec{c}$ be two vectors perpendicular to each other in the xy – plane. If $\vec{r}_i$, $i = 1,2,....n$ are the vectors in the same plane having projection 1 and 2 along b and c respectively $\sum_{i=1}^{n} \left|\vec{r}_i\right|^2$ is equal to k then $\frac{k+20}{3}$ equals to

52. Let f be a continuous function on R such that $f\left(\frac{1}{4n}\right) = \left(\sin e^n\right)e^{-n^2} + \frac{n^2}{n^2+1}$. Then the value of $f(0)$ is

53. If $I = \int_{1/e}^{e} |\log x| \frac{dx}{x^2}$, then $\frac{I}{4} + \frac{1}{2e} =$ _____

54. A bag contain 4 tickets numbered 1, 2, 3, 4 and another bag contains 6 tickets numbered 2, 4, 6, 7, 8, 9. One bag is chosen and a ticket is drawn. The probability that the ticket bears the number 4, is equal to $\frac{A}{B}$ (where A and B are co prime) then the value of $\frac{100A}{B}$ is

Space for Rough work